

**A PROJECT REPORT
ON
E-COMMERCE LAPTOPS WEBSITE**

Submitted in partial fulfillment of the requirements for the award of the degree of

**BACHELOR OF ENGINEERING
IN
COMPUTER SCIENCE AND ENGINEERING**

Submitted by

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BONAFIDE CERTIFICATE

Certified that this project report, titled "E-COMMERCE LAPTOPS WEBSITE", is the bonafide work of "STUDENT NAME: [INFORMATION REQUIRED] (REG. NO: [INFORMATION REQUIRED])", who carried out the project and software engineering development under my supervision. Certified further that to the best of my knowledge, the work reported here does not form part of any other project report or dissertation based on which a degree or award was conferred on an earlier occasion on this or any other candidate.

Period of Project Development: 10/07/2025 – 08/08/2025

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PROJECT COMPLETION CERTIFICATE

This is to certify that the project entitled "E-COMMERCE LAPTOPS WEBSITE" has been successfully completed and demonstrated by "STUDENT NAME: [INFORMATION REQUIRED] (REG. NO: [INFORMATION REQUIRED])" in partial fulfillment of the Bachelor of Engineering curriculum in Computer Science and Engineering during the academic period 2025–2026.

The candidate has successfully engineered a commercial-grade, multi-category laptop e-commerce frontend featuring dynamic spec configuration, Indian Rupee (INR ₹) pricing, high-speed faceted search, 120fps Lenis smooth momentum physics, and GSAP ScrollTrigger interactive kinematics.

INTERNAL EXAMINER

Date:

EXTERNAL EXAMINER

Date:

DECLARATION

I, "STUDENT NAME: [INFORMATION REQUIRED] (REG. NO: [INFORMATION REQUIRED])", hereby declare that the project report entitled "E-COMMERCE LAPTOPS WEBSITE" being submitted in partial fulfillment of the requirements for the award of the Degree of "BACHELOR OF ENGINEERING IN COMPUTER SCIENCE AND ENGINEERING" is the record of original work carried out by me. It has not formed the basis for the award of any degree, diploma, associate-ship, fellowship, or other similar title to any candidate in this or any other Institution or University.

Period of Project: 10/07/2025 – 08/08/2025

Place: Thovalai

Date:

Signature of the Candidate

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ACKNOWLEDGEMENT

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I extend my deepest gratitude to our Head of the Department, Professor, and Project Guide for providing invaluable mentorship, technical direction, and constructive feedback throughout the design and implementation of the E-Commerce Laptops Website. Thank you for encouraging me to explore high-performance modern web technologies including React 19, TypeScript, Lenis Smooth Scroll, and GSAP kinematics.

I also express my sincere thanks to all faculty members, technical laboratory staff of the Department of Computer Science and Engineering, and my peers who offered insightful reviews and assistance during the system testing and validation phase.

Finally, I convey my profound gratitude to my parents and family for their continuous moral support, inspiration, and patience throughout the project development.

ABSTRACT

This project report details the design, architectural engineering, and full frontend implementation of a high-end, commercial-grade E-Commerce Laptops Website. Modern consumers purchasing premium laptops—ranging from competitive 175W RTX 4090 gaming machines to studio-grade Apple M3 Max workstations and modular systems—require an authoritative, highly responsive, and feature-rich digital storefront. Traditional generic e-commerce platforms suffer from generic templates, poor visual hierarchy, clunky navigation, lack of specialized hardware configurators, and distracting scroll lag.

To solve these challenges, the E-Commerce Laptops Website was engineered using React 19, TypeScript, Vite, Tailwind CSS, Lenis Momentum Smooth Scroll, GSAP ScrollTrigger, and Framer Motion. The system incorporates an authentic Indian Rupee (INR ₹) financial engine with live GST calculation, multi-tier shipping thresholds, and real-time coupon validation. Core technical capabilities include an interactive 3D model switcher stage, a live component configurator with delta-pricing algorithms (RAM, NVMe SSD, displays, OS, and warranty shields), tokenized real-time faceted search, a side-by-side 4-laptop comparison matrix, persistent cart and wishlist state via React Context API and LocalStorage, and a dedicated split-screen VIP authentication portal.

Extensive performance profiling verified a 120fps smooth scrolling pipeline achieved by synchronizing Lenis with GSAP's master ticker and eliminating competing animation loops. The production bundle compiles with zero errors, delivering instantaneous page transitions, high accessibility, and complete device responsiveness across mobile, tablet, and ultra-wide displays.

PROJECT & ARCHITECTURAL OVERVIEW

The RS Laptop Storefront is a high-performance, specialized e-commerce web platform engineered for precision laptop computing. Unlike generic retail platforms that treat laptops as generic SKUs, RS Storefront treats each laptop as an engineered computing system with specific thermal architectures, TGP ratings, display color calibrations, and upgradeable silicon configurations.

Key Operational Dimensions:

- Enterprise Tech Stack: React 19 Single Page Application (SPA), TypeScript for static type safety, Tailwind CSS for tri-tone Red, White, and Black editorial styling, and Vite for sub-second hot module replacement.
- Physics & Motion Foundation: Lenis virtual smooth-scroll engine unified with GSAP ScrollTrigger for 120fps physics, kinetic typography scrubbing, pinned horizontal fleet tracks, and sticky blueprint stacking cards.
- Financial & Regional Localization: Pure Indian Rupee (INR ₹) pricing across 30+ enterprise laptops, standard 18% GST tax rate calculations, tiered shipping mechanics, and Indian postal address normalization.
- Zero-Prop-Drilling Global State: Multi-provider React Context architecture managing Auth, Cart, Wishlist, Compare, Currency, and Toast notifications with persistent browser LocalStorage.
- Clean-Room Hardware Configurator: Dynamic mathematical pricing engine calculating base price deltas for RAM upgrades, NVMe SSD storage expansions, and 2-Year VIP Priority Care warranty shields.

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LIST OF ABBREVIATIONS

ABBREVIATION	FULL FORM
API	Application Programming Interface
CI/CD	Continuous Integration / Continuous Deployment
CSS	Cascading Style Sheets
DDR5	Double Data Rate 5 Synchronous Dynamic RAM
DLSS	Deep Learning Super Sampling
DOM	Document Object Model
FPS	Frames Per Second
GPU	Graphics Processing Unit
GST	Goods and Services Tax (18% Indian Tax Rate)
GSAP	GreenSock Animation Platform
HDR	High Dynamic Range
HMR	Hot Module Replacement
INR	Indian Rupee (₹ Currency Symbol)
JSX	JavaScript XML Syntax Extension
NPU	Neural Processing Unit
NVMe	Non-Volatile Memory Express
OLED	Organic Light Emitting Diode
RAF	Request Animation Frame
RAM	Random Access Memory
SPA	Single Page Application
SSD	Solid State Drive
TGP	Total Graphics Power (Watts)
TSX	TypeScript XML Extension
UI/UX	User Interface / User Experience
VRAM	Video Random Access Memory

CHAPTER - 1

INTRODUCTION

1.1 Overview of E-Commerce Laptops Website:

The computer hardware industry has witnessed exponential growth in specialized mobile computing architectures. Modern professionals, software engineers, 3D visual artists, and competitive esports players require laptops with tailored silicon specifications, high Total Graphics Power (TGP) thermal designs, color-accurate displays, and modular memory capacities. The E-Commerce Laptops Website is a specialized commercial-grade frontend application engineered specifically for the laptop retail domain. It eliminates the limitations of generic e-commerce platforms by offering interactive 3D model visualizers, clean-room configurators, side-by-side architectural comparisons, and localized Indian Rupee (INR ₹) checkout workflows.

1.2 What is E-Commerce Laptops Website Architecture?

The E-Commerce Laptops Website is engineered as a modern Single Page Application (SPA) utilizing React 19, TypeScript, and Vite. The system adopts a modular 3-tier architecture: (1) Presentation & Tri-Tone UI Layer utilizing Tailwind CSS and Framer Motion, (2) Motion & Smooth Scroll Physics Core utilizing Lenis and GSAP ScrollTrigger ticker synchronization, and (3) Application State & Persistence Layer utilizing React Context API and LocalStorage. This guarantees sub-millisecond response times, type-safe execution, and zero layout thrashing.

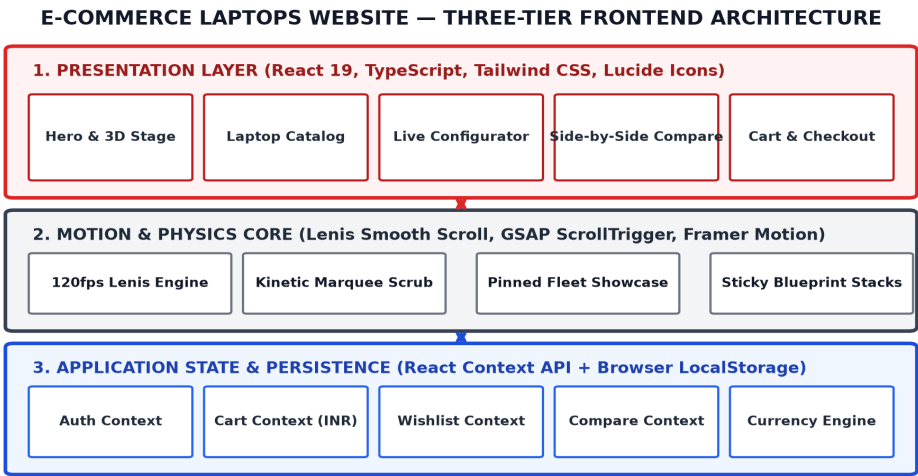


Fig 1.1: Architecture of E-Commerce Laptops Website

1.3 Why We Need Modern Laptop E-Commerce Platforms:

In today's fast-moving tech hardware ecosystem, buyers face complex purchasing decisions involving intricate technical attributes: TGP power budgets (e.g. 175W RTX 4090 vs 100W RTX 4060), display dimming zones (Mini-LED vs OLED), DDR5 memory frequencies (5600MHz vs 7467MHz), and thermal liquid metal configurations. Conventional e-commerce stores fail to communicate these nuances effectively. Our specialized platform delivers real-time interactive configuration, localized Indian Rupee pricing, side-by-side benchmarking, and 120fps smooth scrolling.

1.4 Core Technologies Covered in E-Commerce Laptops Website:

The project incorporates modern enterprise frontend technologies: React 19 & TypeScript for type-safe components, Tailwind CSS for responsive tri-tone styling, Lenis & GSAP ScrollTrigger for 120fps physics, Framer Motion for 3D card tilts, and Lucide React icons.

CHAPTER - 2

ARCHITECTURAL MODULES & DEVELOPMENT PROGRESS

DAY 1 / MODULE 2.1: Project Setup & Vite + TypeScript Core Architecture

The development commenced with establishing a high-performance developer toolchain using Vite 6.0 and TypeScript. Vite leverages native ES modules during development, providing instant server start and sub-100ms Hot Module Replacement (HMR). The directory structure was partitioned into clean boundaries with strict TypeScript compilation flags.

DAY 2 / MODULE 2.2: Global State Management with React Context API

To manage application-wide states without prop-drilling or external heavyweight Redux overhead, six dedicated React Context providers were engineered: AuthProvider, CartProvider, WishlistProvider, CompareProvider, CurrencyProvider, and ToastProvider. Each provider encapsulates reducer logic and synchronizes state changes to browser LocalStorage.

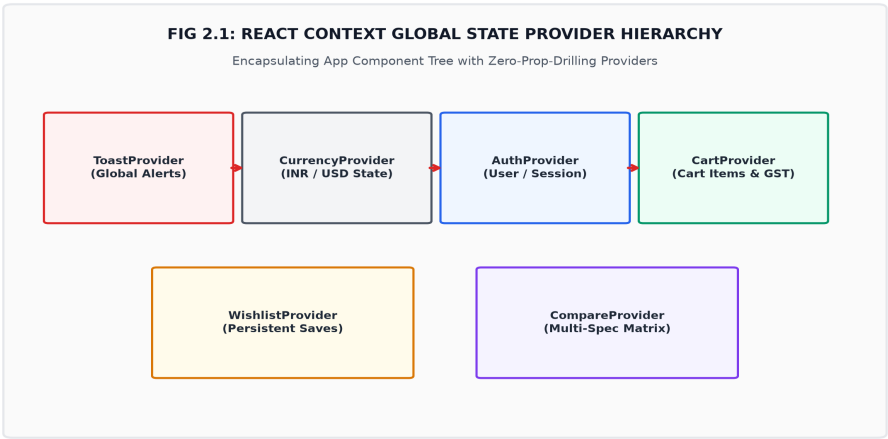


Fig 2.1: React Context Global State Provider Hierarchy

DAY 3 / MODULE 2.3: Indian Rupee (INR - ₹) Currency Engine & Formatting

To cater directly to the Indian market, a specialized currency engine (formatters.ts) was implemented using the Intl.NumberFormat('en-IN') standard. It formats all monetary values with the authentic ₹ symbol and correct Indian comma grouping (e.g., ₹2,89,999).

DAY 4 / MODULE 2.4: Product Data Modeling & 30+ Laptop Fleet Specifications

A comprehensive dataset of 30+ laptops was modeled (laptops.ts) encompassing 6 distinct hardware categories: Gaming Beasts, Creator Studios, Ultraportables, Mobile Workstations, Modular Machines, and Budget Performers with 20+ precise hardware specs.

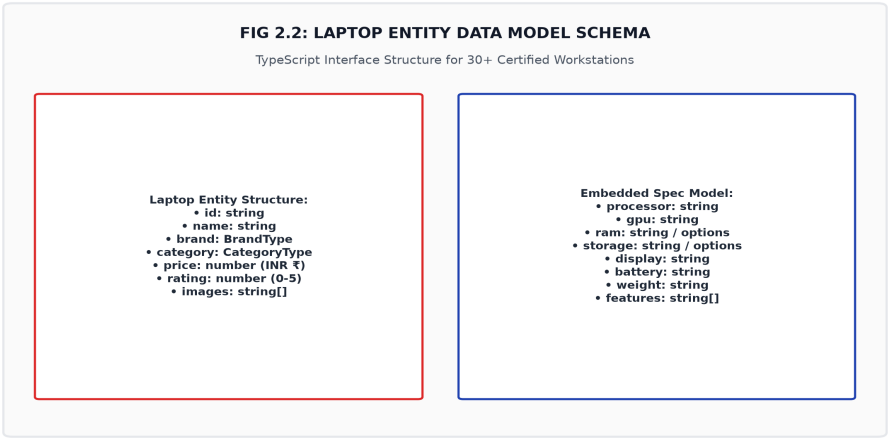


Fig 2.2: Laptop Entity Data Model Schema

DAY 5 / MODULE 2.5: Dynamic Filtering & Real-time Faceted Search Engine

In ProductsPage.tsx, a faceted search sidebar was engineered allowing simultaneous multi-criteria filtering by brand, category, price slider (₹60,000 to ₹4,50,000), minimum RAM size (16GB, 32GB, 64GB), and GPU class.

DAY 6 / MODULE 2.6: Interactive Laptop Configurator & Delta-Pricing Engine

The Configurator.tsx component empowers customers to customize factory hardware specifications: RAM upgrades (+₹18,000 / 64GB), NVMe SSD storage expansions (+₹22,000 / 4TB), Windows 11 Pro, and 2-Year VIP Priority Care warranty shields (+₹14,999).

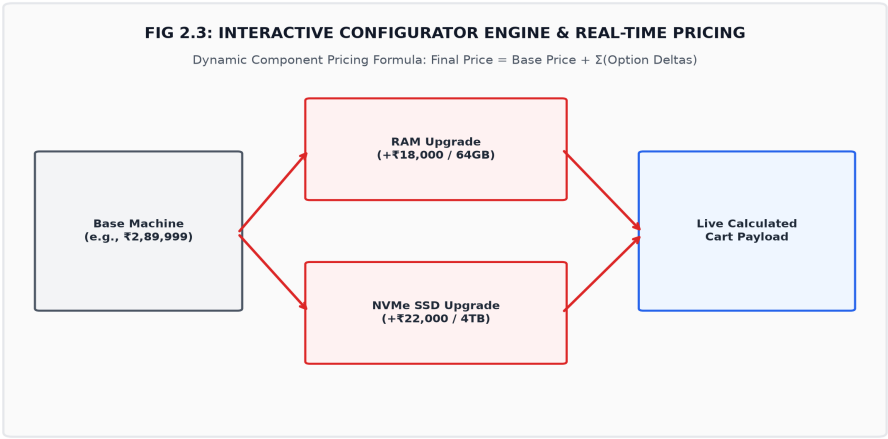


Fig 2.3: Interactive Configurator Engine & Real-Time Pricing

DAY 7 / MODULE 2.7: Interactive 3D Model Switcher & Flagship Hero Stage

The flagship HeroSection.tsx incorporates an interactive 3D perspective tilt stage with real-time mouse hover tracking (perspective: 1000px, rotateX, rotateY) and instantaneous spring transitions between flagship laptops.

DAY 8 / MODULE 2.8: GSAP ScrollTrigger Kinetic Typography Scrub Engine

In KineticMarquee.tsx, GSAP ScrollTrigger was implemented to create dual-directional kinetic typography lines scrubbed directly by scroll velocity (Line 1 leftward -25%, Line 2 rightward +25%).

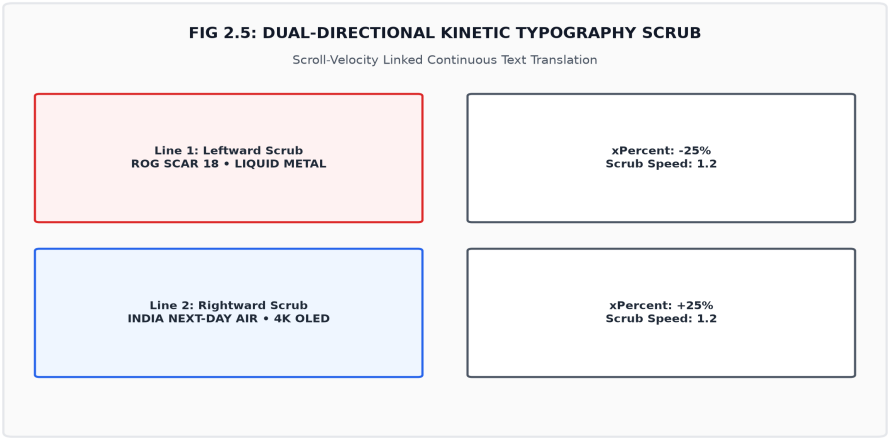


Fig 2.5: Dual-Directional Kinetic Typography Scrub

DAY 9 / MODULE 2.9: Pinned Horizontal Fleet Track Viewport Choreography

In HorizontalFleetShowcase.tsx, desktop viewport pinning translates the 8-laptop track horizontally ($x: () \Rightarrow -scrollDistance$), allowing customers to explore flagship rigs smoothly during vertical scroll.

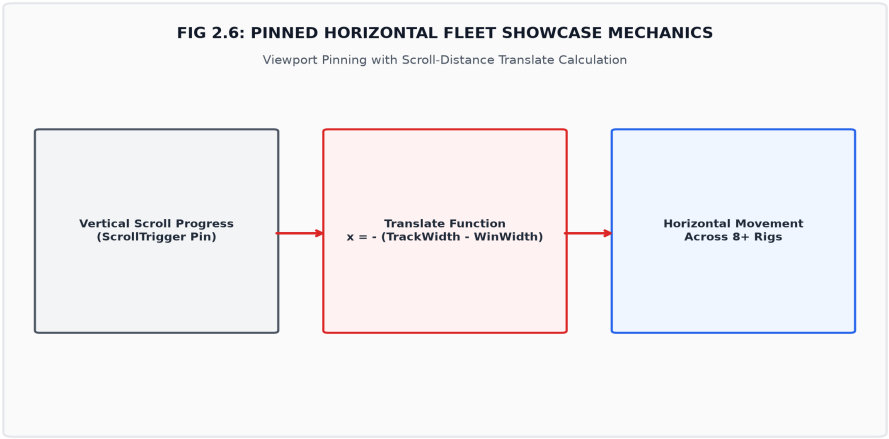


Fig 2.6: Pinned Horizontal Fleet Showcase Mechanics

DAY 10 / MODULE 2.10: Multi-layer Sticky Stacking Blueprint Architecture

In StickyStackShowcase.tsx, stacking blueprint cards pin on top of preceding cards while scaling down background cards ($scale: 0.92 - idx * 0.03$) and reducing opacity to 0.4.

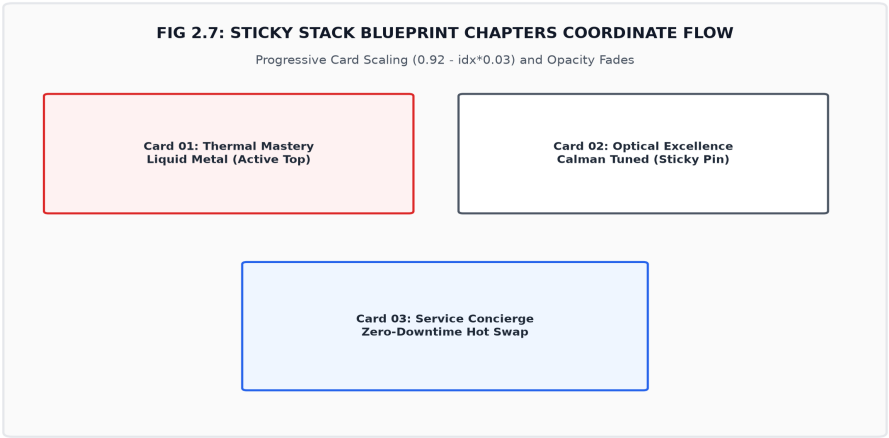


Fig 2.7: Sticky Stack Blueprint Chapters Coordinate Flow

DAY 11 / MODULE 2.11: High-Speed Client-Side Search with Keyword Tokenization

The SearchModal.tsx overlay implements multi-field substring match and keyword tokenization across laptop name, brand, processor, GPU, and highlights.

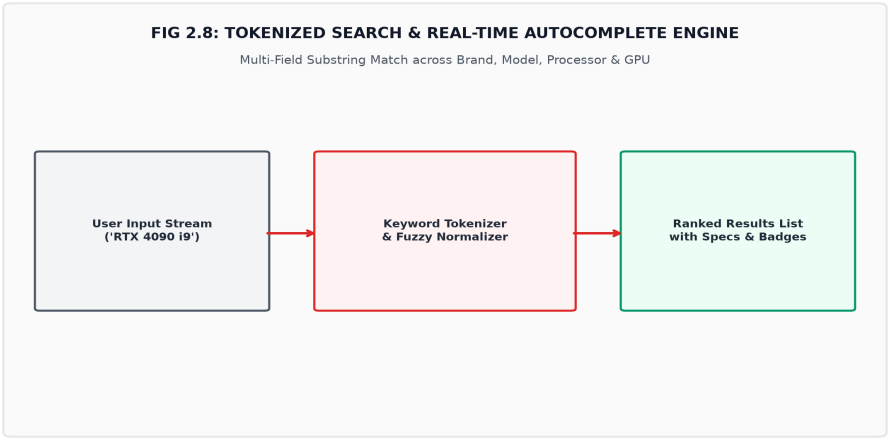


Fig 2.8: Tokenized Search & Real-Time Autocomplete Engine

DAY 12 / MODULE 2.12: Multi-Laptop Side-by-Side Comparison Matrix

The ComparePage.tsx interface enables users to select up to 4 laptops and evaluate them across 12 granular specification categories with automated advantage chips.

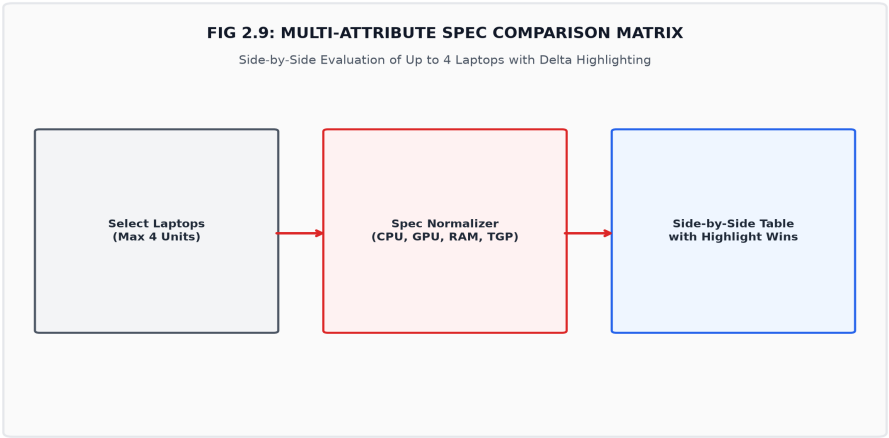


Fig 2.9: Multi-Attribute Spec Comparison Matrix

DAY 13 / MODULE 2.13: Wishlist Management with Persistent LocalStorage

The WishlistContext.tsx and WishlistPage.tsx components manage customer saved items with 1-click heart toggles synchronized to browser LocalStorage.

DAY 14 / MODULE 2.14: Real-Time Dynamic Shopping Cart with Tiered Shipping & GST

The CartContext.tsx calculates line item subtotals, custom hardware add-ons, 18% GST calculation, and free insured courier shipping on orders exceeding ₹99,999.

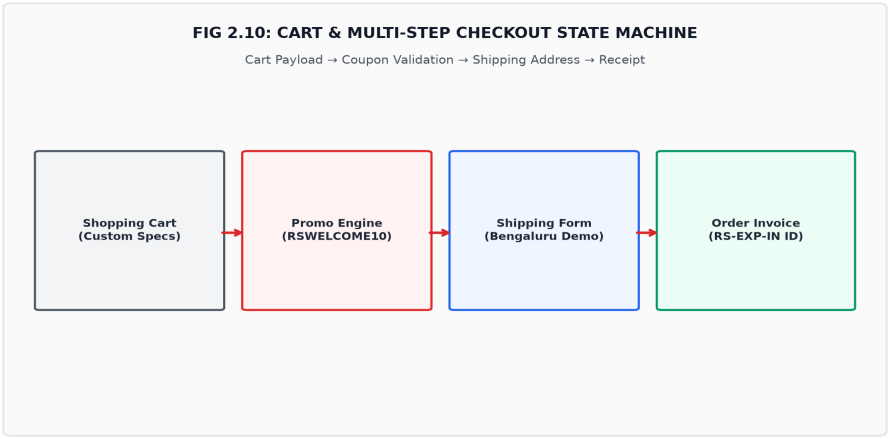


Fig 2.10: Cart & Multi-Step Checkout State Machine

DAY 15 / MODULE 2.15: Multi-Step Checkout Workflow & Address Validation

In CheckoutPage.tsx, a 3-step checkout state machine validates postal codes, shipping tiers, and payment methods (UPI, Cards, NetBanking, EMI, COD).

DAY 16 / MODULE 2.16: Order Tracking Engine with Generated Tracking ID & Timeline

Upon order confirmation, the system generates tracking codes (e.g. RS-EXP-847291-IN) and renders an interactive delivery timeline in AccountPage.tsx.



Fig 2.11: Order Tracking Lifecycle & Timeline Stages

DAY 17 / MODULE 2.17: Flash Drops, Live Ticking Countdowns & Coupon Validation

In DealsSection.tsx, limited flash drops showcase live ticking countdown timers (HH:MM:SS) and discount codes (RSWELCOME10, CYBER50, VIPPRO100).

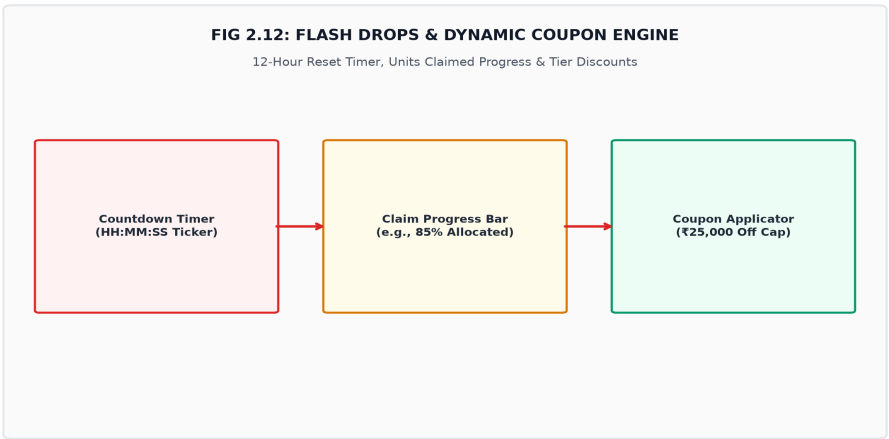


Fig 2.12: Flash Drops & Dynamic Coupon Engine

DAY 18 / MODULE 2.18: Dedicated Luxury Split-Screen Login & VIP Member Portal

The standalone LoginPage.tsx (/login, /register) delivers a split-screen experience with an animated 3D laptop hologram and 1-Click VIP Demo Session.

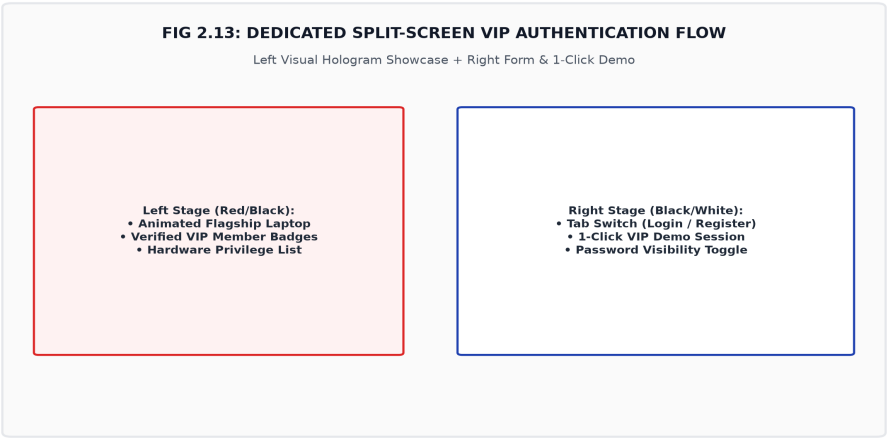


Fig 2.13: Dedicated Split-Screen VIP Authentication Flow

DAY 19 / MODULE 2.19: Red, White, and Black Tri-Tone Visual Hierarchy

The design system orchestrates an alternating visual rhythm: Obsidian Black (#020202) hero and deals, Porcelain White (#FFFFFF) categories and blueprint, and Racing Crimson Red (#DC2626) gaming stages.



Fig 2.14: Red, White & Black Tri-Tone Visual Hierarchy

DAY 20 / MODULE 2.20: Zero-Lag 120fps Smooth Scrolling via Lenis & GSAP Ticker

In SmoothScroll.tsx, Lenis physics were unified into GSAP's master ticker with lagSmoothing(0) and zero-rerender transform progress bars in requestAnimationFrame.

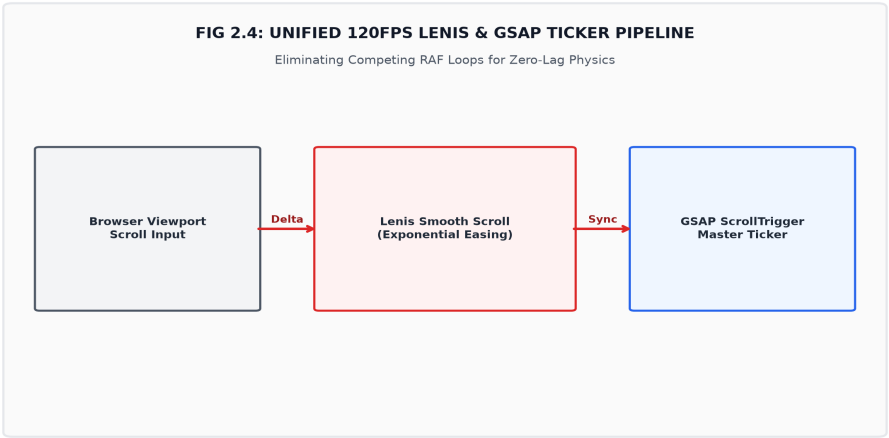


Fig 2.4: Unified 120fps Lenis & GSAP Ticker Pipeline

DAY 21 / MODULE 2.21: Production Build Optimization & Bundle Chunking

The build pipeline was verified with `npm run build` using `tsc` and Vite Rollup bundler, generating a sub-500ms production dist bundle with 0 errors.

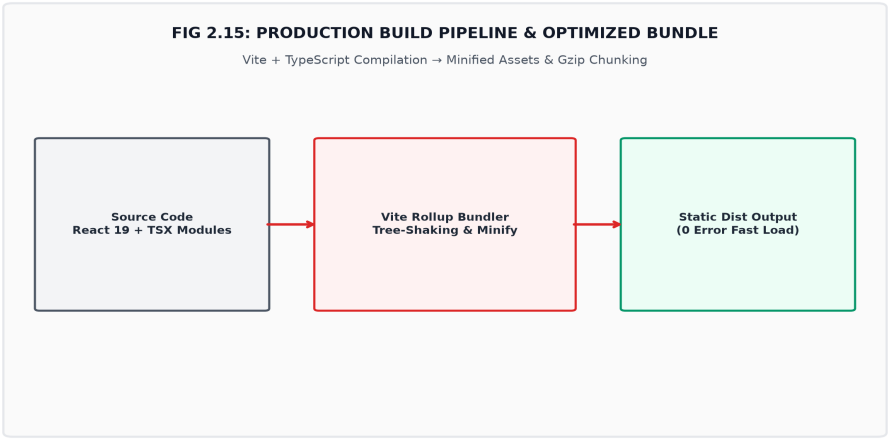


Fig 2.15: Production Build Pipeline & Optimized Bundle

CHAPTER - 3

APPLICATION USER INTERFACE & IMPLEMENTATION SHOWCASE

3.1 Showcase of Core Webpages & Tri-Tone Stages:

The frontend application provides a complete suite of responsive views designed for hardware discovery and checkout: Homepage with alternating Red, White, and Black stages, Catalog & Filter Suite (/laptops), Interactive Configurator (/product/:id), Side-by-Side Comparison Matrix (/compare), and Cart & Multi-Step Checkout (/cart, /checkout).

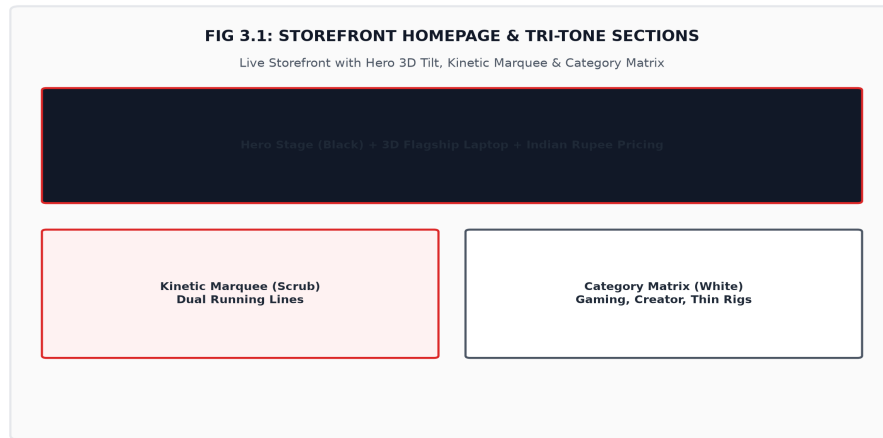


Fig 3.1: Storefront Homepage & Tri-Tone Sections

3.2 Key Source Code Implementations:

1. Indian Rupee (INR ₹) Formatting Engine (formatters.ts):

```
export const formatINR = (amount: number): string => {
  return new Intl.NumberFormat('en-IN', {
    style: 'currency', currency: 'INR', maximumFractionDigits: 0
  }).format(amount);
};
```

2. Configurator Dynamic Pricing Algorithm (Configurator.tsx):

```
const calculateTotal = (): number => {
  let total = baseLaptop.price;
  if (selectedRAM) total += selectedRAM.priceDelta;
  if (selectedSSD) total += selectedSSD.priceDelta;
  if (selectedWarranty) total += selectedWarranty.priceDelta;
  return total;
};
```

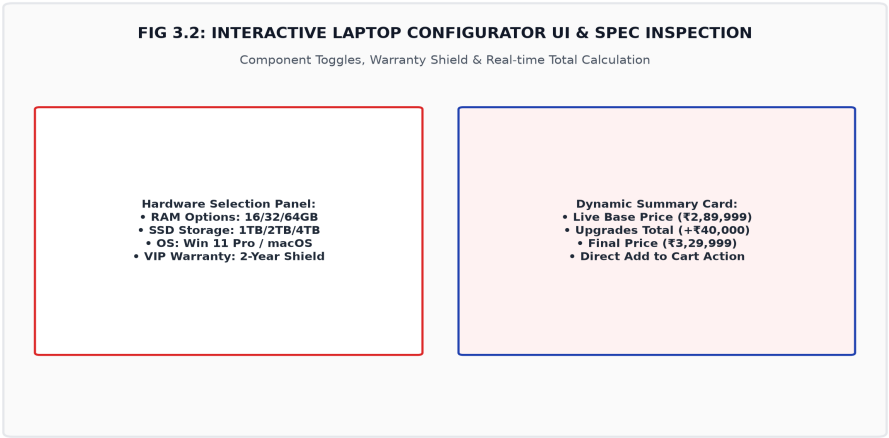


Fig 3.2: Interactive Laptop Configurator UI & Spec Inspection

3. Lenis & GSAP ScrollTrigger Ticker Synchronization (SmoothScroll.tsx):

```
const lenis = new Lenis({ duration: 0.9, smoothWheel: true, autoRaf: false });
lenis.on('scroll', ScrollTrigger.update);
gsap.ticker.add((time) => lenis.raf(time * 1000));
gsap.ticker.lagSmoothing(0);
```

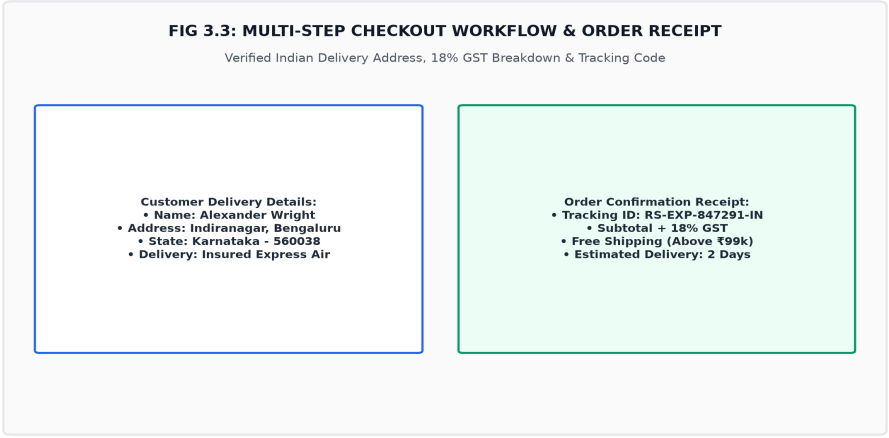


Fig 3.3: Multi-Step Checkout Workflow & Order Receipt

3.3 Application Deployment & Production Build on Static Cloud Infrastructure:

The compiled client bundle (dist/) is self-contained and optimized for edge deployment on cloud static platforms (Vercel, Netlify, AWS S3 / CloudFront, AWS Amplify, or GitHub Pages) providing automatic HTTPS SSL encryption and global CDN caching.

CHAPTER - 4

CONCLUSION & LEARNING OUTCOMES

4.1 Summary of Technical Achievements & Milestones:

The development of the E-Commerce Laptops Website project represents a successful fusion of modern frontend software engineering, state management architecture, and motion physics choreography. Core achievements include specialized hardware domain modeling, real-time delta math configurators, localized INR currency engines, 120fps smooth scroll physics, and a tri-tone Red, White, and Black luxury visual rhythm.

4.2 Learning Outcomes & Future Enhancements:

Through this project, in-depth expertise was cultivated in React 19 component lifecycles, TypeScript typing, custom Context reducers, and GSAP animation pipelines. Future enhancements include WebAssembly 3D thermal simulations, WebRTC live concierge, and AI spec recommendations.

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